Q1 JEE Main 2020 - 3 September (Morning)

An element with molar mass $2.7 \times 10^{-2} \mathrm{kg mol}^{-1}$ forms a cubic unit cell with edge length 405 pm. If its density is $2.7 \times 10^{3} \mathrm{kg m}^{-3}$, the radius of the element is approximately $\times 10^{-12} \mathrm{m}$ (to the nearest integer).

Q2 JEE Main 2020 - 5 September (Morning)

A diatomic molecule X_2 has a body-centred cubic (bcc) structure with a cell edge of $300 \mathrm{pm}$. The density of the molecule is $6.17 \mathrm{g\ cm}^{-3}$. The number of molecules present in $200 \mathrm{g}$ of X_2 is :
(Avogadro constant $\left(\mathrm{N_A}\right) = 6 \times 10^{23} \mathrm{mol}^{-1}\right)$

(A) $40 \mathrm{N_A}$
(B) $4 \mathrm{N_A}$
(C) $2 \mathrm{N_A}$
(D) $8 \mathrm{N_A}$

Q3 JEE Main 2020 - 5 September (Evening)

An element crystallises in a face-centred cubic (fcc) unit cell with cell edge a. The distance between the centres of two nearest octahedral voids in the crystal lattice is

(A) a
(B) $\sqrt{2}$ a
(C) $\frac{\mathrm{a}}{\sqrt{2}}$
(D) $\frac{a}{2}$

Q4 JEE Main 2020 - 6 September (Evening)

A crystal is made up of metal ions ' M_1' and 'M_2' and oxide ions. Oxide ions form a ccp lattice structure. The cation 'M_1' oc cupies 50% of octahedral voids and the cation 'M_2' occupies 12.5% of tetrahedral voids of oxide lattice. The oxidation numbers of 'M_1' and 'M_2' are, respectively.

(A) $+3, +1$
(B) $+4, +2$
(C) $+1, +3$
(D) $+2, +4$

Q5 JEE Main 2020 - 8 January (Evening)

Which of the following compounds is likely to show both Frenkel and Schottky defects in its crystalline form?

(A) $AgBr$
(B) KCl
(C) $CsCl$
(D) ZnS

Answer Key

Q1 (143) **Q2** (B) **Q3** (C) **Q4** (D)
Q5 (A)

Q1

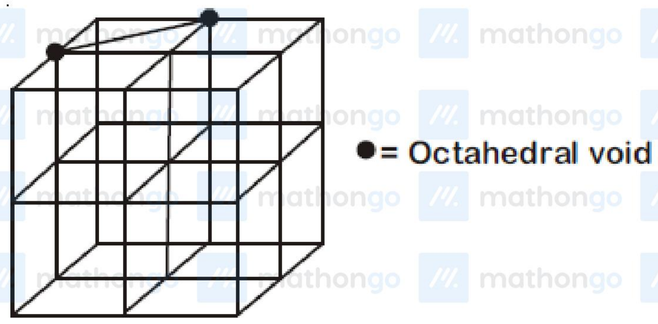
M, Molar mass of an element $= 27\mathrm{gmmol}^{-1}$ a, edge length of a cubic unit cell = $405\mathrm{pm}$
$= 4.05 \times 10^{-8}\mathrm{cm}$
d, density of the element $= 2.7\mathrm{gm/cc}$
$\mathrm{d} = \frac{\mathrm{Z} \times \mathrm{M}}{\mathrm{N_A} \times (\mathrm{a})^3}$
$2.7 = \frac{\mathrm{Z} \times 27}{6 \times 10^{23}\left(4.05 \times 10^{-8}\right)^3}; \mathrm{Z} = 4$
The element has fce unit cell
$\therefore \quad \sqrt{2}\mathrm{a} = 4\mathrm{r}; \quad \mathrm{r} = \frac{1.414 \times 405}{4} \simeq 143\mathrm{pm}$
$= 143 \times 10^{-12}\mathrm{m}$

Q2

$\mathrm{d} = \frac{\mathrm{Z} \times \mathrm{M}}{\mathrm{a}^3 \times \mathrm{N_A}}$
$6.17 = \frac{2 \times \mathrm{M}}{\left(3 \times 10^{-8}\right)^3 \times 6 \times 10^{23}}$ (For $\mathrm{BCC} \mathrm{Z} = 2)$
$\mathrm{M} = 50\mathrm{g/mol}$
in $200\mathrm{g}$, 4 moles of $\mathrm{X_2}$ is present

Q3

Octahedral void in fcc is present at edge centre and body centre.



minimum distance between two nearest
octahedral void is $= \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2} = \frac{a}{\sqrt{2}}$

Q4

$\mathrm{M}_1 \quad \mathrm{M}_2 \quad \mathrm{O}$

$50\% \quad 12.5\% \quad \mathrm{cCp}$

octahedral tetrahedral void void

$\frac{50}{100} \times 4 \quad \frac{12.5}{100} \times 8$

Charge $2x \quad 1y \quad 4 \times 2$

$2x + y = 8$

$x = 2$

$y = 4$

Q5

Only $AgBr$ can exhibit both Schottky and Frenkel defect